
Project Update Report

Overview

This report presents my individual progress and technical contributions to the Constraint Satisfaction Problem (CSP)-based Sudoku Solver project. My assigned area of work has been focused on developing the search mechanism of the solver. I have been responsible for investigating backtracking techniques, designing the solving approach, and building the initial framework required to perform automated Sudoku solving. The current work mainly covers algorithm research and early implementation, while optimization and complete integration are still in progress.

Week 1 (June 24 – June 30): Backtracking Research and Algorithm Analysis

- **Backtracking Algorithm Study:** I began by researching the fundamentals of the backtracking search algorithm and its application to Constraint Satisfaction Problems (CSPs). The focus was on understanding how recursive exploration works, how decisions are made during the search process, and how incorrect choices are reversed.
- **Complexity Investigation:** I analyzed the time and space complexity of applying a naive backtracking approach to a standard 9×9 Sudoku puzzle. This research helped identify the limitations of basic search methods and the need for optimization strategies to improve solving efficiency.
- **Initial Planning:** Based on the research, I outlined the structure of the backtracking solver and identified the key stages required, including variable selection, value assignment, constraint checking, and backtracking.

Week 2 (July 1 – July 7): Backtracking Algorithm Design and Framework Preparation

- **Pseudocode Development:** I created detailed pseudocode for the backtracking search algorithm. The design describes how the solver should:
 - Select an unassigned Sudoku cell.
 - Try possible values from the available domain.
 - Check whether the assignment satisfies Sudoku constraints.
 - Return to previous decisions when a conflict occurs.
- **Implementation Structure:** I started preparing the function structure for the backtracking algorithm. The initial framework was designed to allow future integration with the CSP constraint-checking system.
- **Testing Preparation:** I reviewed possible testing approaches to ensure that the algorithm could later be evaluated using different Sudoku difficulty levels.

Week 3 (July 8 – July 15): Initial Backtracking Implementation

- **Backtracking Function Development:** I began implementing the main recursive backtracking function. The current implementation establishes the basic search flow and provides the foundation for solving Sudoku puzzles.
 - **Constraint Integration Preparation:** I started connecting the backtracking structure with the constraint-checking functions developed by the CSP module. At this stage, the integration is still being refined.
 - **MRV Heuristic Preparation:** I reviewed the Minimum Remaining Values (MRV) heuristic and began preparing the algorithm structure for future integration. The purpose of this heuristic is to improve search efficiency by selecting the variable with the smallest number of possible values first.
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